Introduction to IP AddressingTheory

1.List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

ANS:

Differences Between IPv4 and IPv6

IPv4	**IPv6**
1. Uses 32-bit addresses.	1. Uses 128-bit addresses.
2. Written in decimal format, e.g., 192.168.1.1.	2. Written in hexadecimal format, e.g., 2001:db8::1.
3. Provides about 4.3 billion addresses.	3. Provides a vastly larger number of addresses.
4. Uses subnet masks, e.g., 255.255.255.0.	4. Uses prefix lengths, e.g., /64.
5. Supports broadcast communication.	5. Uses multicast and anycast; it does not use broadcast.
6. Often uses NAT to share public IPv4 addresses.	6. Generally reduces the need for NAT because of its large address space.

2.Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

ANS:

Static IPv4 Address Assignment for Home Wi-Fi

Network: 192.168.10.0/24

Device	**IP Address**	**Subnet Mask**	**Default Gateway**
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Smartphone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

Explanation:

- IP Address: A unique address assigned to each device.
- Subnet Mask: 255.255.255.0 identifies the local network.
- Default Gateway: 192.168.10.1 is the router that forwards traffic to other networks.

3.Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

ANS:

Subnetting Calculation: 192.168.1.0/27

Given:

- Network: 192.168.1.0/24
- New subnet mask: /27 = 255.255.255.224

Calculation:

- Bits borrowed: \(27-24=3\)
- Number of subnets: \(2^3=8\)
- Host bits remaining: \(32-27=5\)
- Addresses per subnet: \(2^5=32\)
- Usable hosts per subnet: \(32-2=30\)

First Three Subnets

Subnet	Network Address	Broadcast Address	Usable Host IP Range
1	192.168.1.0	192.168.1.31	192.168.1.1 – 192.168.1.30
2	192.168.1.32	192.168.1.63	192.168.1.33 – 192.168.1.62
3	192.168.1.64	192.168.1.95	192.168.1.65 – 192.168.1.94

Final Answer:

- Total subnets: 8
- Usable hosts per subnet: 30
- Subnet block size: 32 addresses

**4.You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.**

ANS:

Given Information

- IP Address: 10.0.5.17
- Subnet Mask: 255.255.255.0
- CIDR Notation: /24

Calculations

1. Network Address: 10.0.5.0
2. Broadcast Address: 10.0.5.255
3. Valid Host Address Range: 10.0.5.1 to 10.0.5.254
4. Total Usable Hosts: \(2^8-2=254\)

Final Answer

Item	Result
Network Address	10.0.5.0
Broadcast Address	10.0.5.255
First Valid Host	10.0.5.1
Last Valid Host	10.0.5.254
Usable Hosts	254

5.Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

ANS:

Subnet Calculator Verification

Steps:

1. Open the Online Subnet Calculator.
2. Enter IP address: 10.0.5.17
3. Enter subnet mask: 255.255.255.0 (/24).
4. Click Calculate and take a screenshot of the results.

Expected Results

Field	Calculator Output

IP Address	10.0.5.17
CIDR	/24
Network Address	10.0.5.0
Broadcast Address	10.0.5.255
First Usable Host	10.0.5.1
Last Usable Host	10.0.5.254
Usable Hosts	254

These are the expected results for a /24 subnet.